Analog Electronic Circuits
Quadrature Down Converter Project

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***Abstract*—This report addresses the fundamental need for Quadrature (I-Q) down-conversion in modern wireless communication systems. While standard single-branch mixing effectively translates high-frequency Radio Frequency (RF) signals to a lower Intermediate Frequency (IF), it suffers from a fatal flaw known as the Image Frequency problem. By mathematically analyzing the mixing process, this paper demonstrates how splitting the local oscillator into two orthogonal phases (In-Phase and Quadrature) preserves the phase polarity of the frequency difference. This orthogonal separation allows modern receivers to cancel image interference, double spectral efficiency, and process complex modulation schemes such as QAM and PSK. This report details the complete design, simulation, and hardware realization of a QDC system including the oscillator, mixer, filters, and amplifiers.**

***Index Terms*—Quadrature Down Converter, Mixer, Intermediate Frequency, Low Pass Filter, Common-Source Amplifier, Oscillator.**

I. Introduction: The Need for Quadrature Mixing (Q1)

A. The Flaw of Standard Mixing

In a traditional heterodyne receiver, an incoming RF signal (ω_{RF}) is multiplied by a local oscillator (ω_{LO}) to step the frequency down to a manageable Intermediate Frequency (ω_{IF}). The mathematical operation of a standard mixer is a simple trigonometric multiplication:

$$\cos(\omega_{RF}t) \times \cos(\omega_{LO}t) = \frac{1}{2}\left[\cos((\omega_{RF}-\omega_{LO})t) + \cos((\omega_{RF}+\omega_{LO})t)\right] \quad (1)$$

A low-pass filter easily removes the high-frequency sum, leaving only the difference $|\omega_{RF} - \omega_{LO}|$. However, the absolute value function reveals a critical physical vulnerability. Suppose our desired signal is located at $\omega_{LO}+\omega_{IF}$. The mixer will output ω_{IF}.

Now suppose there is an unwanted noise signal, or "Image," located at $\omega_{LO} - \omega_{IF}$. Because cosine is an even function ($\cos(-x) = \cos(x)$), the mixer will process the image as follows:

$$\cos(-\omega_{IF}t) = \cos(\omega_{IF}t) \quad (2)$$

Both the desired signal and the unwanted image noise map into the exact same IF band, permanently corrupting the data. A standard single-branch mixer cannot distinguish between frequencies that lie above the LO and frequencies that lie below the LO.

B. The Quadrature (I-Q) Solution

To solve this, modern receivers utilize a Quadrature Down Converter (QDC). Instead of one mixer, the RF signal is split and fed into two parallel mixers. The LO signal is also split; one branch remains untouched (In-Phase, or I), while the other is shifted by exactly 90° (Quadrature, or Q).

Let the Local Oscillator signals be:

$$LO_I = \cos(\omega_{LO}t) \quad (3)$$

$$LO_Q = -\sin(\omega_{LO}t) \quad (4)$$

1) Processing the Desired Signal: Let the desired signal be higher than the LO: $\omega_{RF} = \omega_{LO} + \omega_{IF}$.

$$I_{desired} = \cos((\omega_{LO}+\omega_{IF})t)\cos(\omega_{LO}t) \xrightarrow{\text{LPF}} \frac{1}{2}\cos(\omega_{IF}t) \quad (5)$$

$$Q_{desired} = \cos((\omega_{LO}+\omega_{IF})t)(-\sin(\omega_{LO}t)) \xrightarrow{\text{LPF}} \frac{1}{2}\sin(\omega_{IF}t) \quad (6)$$

2) Processing the Image Signal: Now, let the unwanted image signal be lower than the LO: $\omega_{IM} = \omega_{LO} - \omega_{IF}$.

$$I_{image} = \cos((\omega_{LO}-\omega_{IF})t)\cos(\omega_{LO}t) \xrightarrow{\text{LPF}} \frac{1}{2}\cos(-\omega_{IF}t) = \frac{1}{2}\cos(\omega_{IF}t) \quad (7)$$

$$Q_{image} = \cos((\omega_{LO}-\omega_{IF})t)(-\sin(\omega_{LO}t)) \xrightarrow{\text{LPF}} \frac{1}{2}\sin(-\omega_{IF}t) = -\frac{1}{2}\sin(\omega_{IF}t) \quad (8)$$

3) The Mathematical Takeaway: Looking closely at the results, the I channel outputs are identical for both the desired signal and the image. However, the Q channel outputs have **opposite signs**. By using an orthogonal (90-degree) reference, the QDC preserves the polarity of the frequency difference. Because the image and the desired signal have different phase relationships in the Q channel, the Digital Signal Processor (DSP) can mathematically add or subtract the I and Q streams (often via a Weaver or Hartley architecture) to completely cancel out the image frequency, leaving only the pristine desired signal.

C. Utility in Modern Wireless Receivers

The mathematical separation provided by I-Q mixing is the foundational bedrock of modern wireless standards (Bluetooth, Wi-Fi, 5G). Its utility extends beyond simply rejecting image noise:

1) **Direct-Conversion (Zero-IF) Receivers:** By setting the LO frequency exactly equal to the RF frequency ($\omega_{IF} = 0$), the QDC translates the signal directly to baseband. Because $f_{IF} = 0$, the image frequency is the signal itself. I-Q mixing is strictly required here to separate the upper and lower sidebands of the original RF spectrum, preventing the signal from folding over and destroying itself.
2) **Complex Modulation Schemes:** Modern data transmission does not just alter amplitude or frequency; it alters both simultaneously to pack more bits into the same bandwidth. Quadrature Amplitude Modulation (QAM) and Phase-Shift Keying (PSK) rely entirely on I-Q independence. By transmitting unique data on the I carrier and completely independent data on the Q carrier, the spectral efficiency of the communication link is mathematically doubled.

D. Graphical Spectrum Representation

While the time-domain trigonometric equations prove that the Q channel preserves the sign of the frequency difference, looking at the frequency domain (the spectrum) provides a much clearer visual intuition.

If we look at a standard single-mixer spectrum, both the desired RF band and the Image band fold over the LO frequency and land on top of each other at baseband (the IF). Because we only have one axis of information (amplitude), they are permanently superimposed.

However, when we introduce the Q channel, we are effectively adding a second, orthogonal dimension to our spectrum (a complex plane). In this complex frequency domain, positive frequencies (the desired signal) rotate counter-clockwise, while negative frequencies (the image) rotate clockwise. By having both I (real) and Q (imaginary) components, the receiver's digital signal processor can "see" this rotation direction and filter out the image spectrum completely before demodulation.

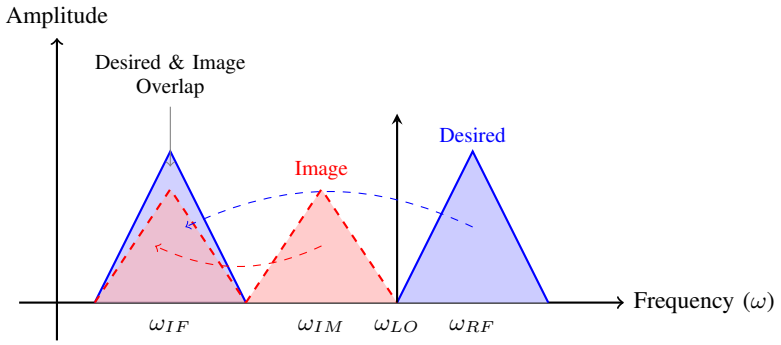


Fig. 1. Spectrum depiction of a standard mixer: The Image and Desired signals irreversibly overlap at the Intermediate Frequency, causing data corruption.

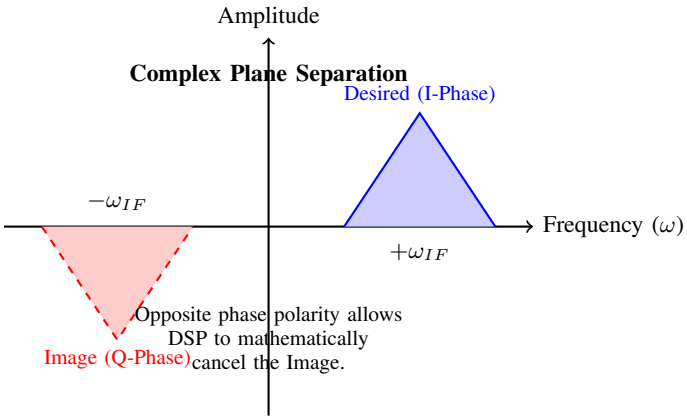


Fig. 2. Spectrum depiction of I-Q mixing: By utilizing phase information, the image spectrum is shifted into the negative complex plane, preserving the desired signal.

E. The Complex Envelope (Euler's Formula)

To tie the physical QDC circuit to the digital software that processes its output, we look to Euler's formula:

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t) \tag{9}$$

In modern digital receivers, the processor treats the I channel as the "Real" part and the Q channel as the "Imaginary" part of a single, complex number stream. The combined baseband signal $S(t)$ can be written as:

$$S(t) = I(t) + jQ(t) \tag{10}$$

If we only had $I(t)$ (a standard mixer), we would only have $\cos(\omega t)$. But because we have $I(t) + jQ(t)$, we possess the full complex exponential $e^{j\omega t}$. Possessing the full complex signal gives the receiver absolute knowledge of the signal's instantaneous amplitude, phase, and frequency trajectory at any given microsecond. This mathematical elegance is the ultimate reason why Quadrature Down Converters are the undisputed standard in every modern wireless device.

I. Quadrature Oscillator Design (Q2)

A quadrature oscillator is a circuit that generates two sinusoidal signals of the same frequency but with a $90°$ phase difference. These are commonly referred to as the Sine (v_{OSCQ}) and Cosine (v_{OSCI}) signals. In this project, we designed a 3-opamp oscillator to meet the target frequency requirements using UA741 op-amps.

A. Circuit Topology and Operation

Our design, as shown in the LTSpice schematic (Fig. 1), utilizes a loop consisting of two integrators and a feedback/limiter stage.

1) **Integrator Stages (U1 and U2):** Each integrator provides a $90°$ phase shift. By cascading two of them, we get a total of $180°$ shift.
2) **The Limiter Stage (U3):** A major problem with oscillators is that the amplitude can grow until it clips against the power rails, turning the sine wave into a square wave. To prevent this, we used a third op-amp stage with 1N4148 diodes in the feedback path (R_8, R_9, R_{10}). These diodes act as a "soft-clipper," stabilizing the amplitude at a level where the waveform remains sinusoidal.
3) **Barkhausen Criterion:** For the circuit to oscillate, the total phase shift around the loop must be $360°$ (or $0°$) and the loop gain must be exactly 1. The third op-amp provides the final $180°$ inversion to satisfy this condition.

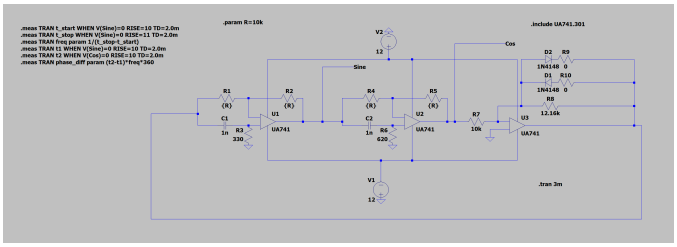


Fig. 1. LTSpice Schematic of the 3-OpAmp Quadrature Oscillator using UA741.

B. Mathematical Derivation

The transfer function of an ideal inverting integrator is given by:

$$H(s) = -\frac{1}{sRC} \tag{1}$$

In the frequency domain ($s = j\omega$):

$$H(j\omega) = -\frac{1}{j\omega RC} = \frac{j}{\omega RC} \tag{2}$$

The presence of the 'j' term indicates a $90°$ phase shift relative to the input.

If the output of the first integrator (U1) is V_{Sine}, and it is fed into the second integrator (U2), the output V_{Cos} will be:

$$V_{Cos} = V_{Sine} \times \left(\frac{1}{j\omega RC}\right) \tag{3}$$

This ensures that V_{Cos} and V_{Sine} are in quadrature. The theoretical frequency of oscillation is determined by:

$$f_{osc} = \frac{1}{2\pi RC} \tag{4}$$

Using our component values of $R = 10k\Omega$ and $C = 1nF$, the ideal frequency would be approximately $15.9kHz$. However, in our simulation, we observe a much higher frequency ($\approx 170kHz$). This shift is primarily due to the non-ideal behavior of the UA741 op-amp (specifically its limited Gain-Bandwidth Product and internal phase shifts) at high frequencies.

C. Simulation Results and Analysis

We performed a transient analysis for $3ms$ to allow the oscillations to settle. The results were captured using LTSpice's '.meas' directives and FFT analysis.

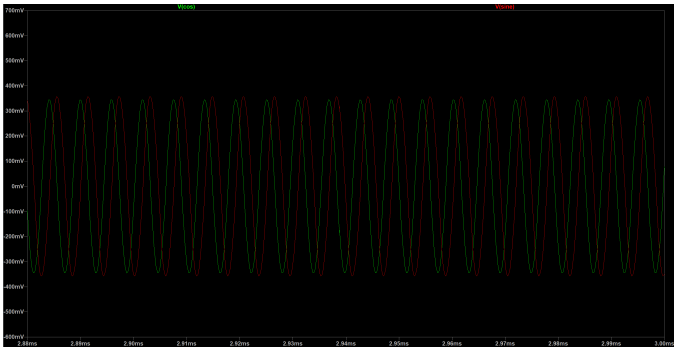


Fig. 2. Transient waveform showing the quadrature Sine (red) and Cosine (green) outputs.

1) Frequency Measurement: As shown in the Spice Error Log (Fig. 3), the measured frequency was $170.76kHz$. While this differs from the simple $1/2\pi RC$ calculation, it is stable and provides a strong "carrier" signal for our mixer stage.

2) Phase Difference: The most critical requirement is the $90°$ phase shift. Using the time-delay measurement between the zero-crossings of the two waves:

$$Phase_Diff = (t_2 - t_1) \times f \times 360° \tag{5}$$

Our simulation yielded a phase difference of $-93.38°$ (Fig. 3). This is extremely close to the ideal $90°$.

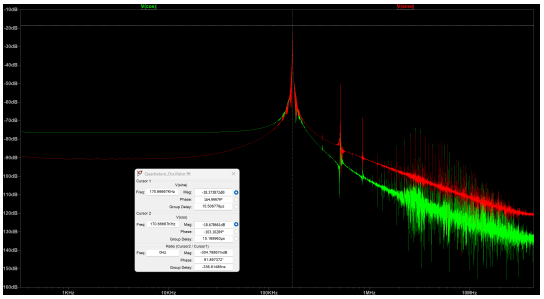


Fig. 3. LTSpice measurement log showing $f = 170.76kHz$ and Phase Difference $\approx -93.4°$.

3) Spectral Purity (FFT): The FFT plot (Fig. 4) shows a sharp peak at the fundamental frequency of $170.8kHz$. The fundamental remains dominant enough for effective mixing.

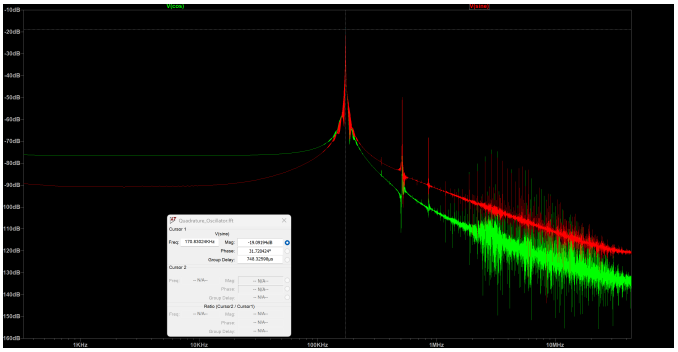


Fig. 4. FFT of the oscillator outputs showing the peak at $170.8kHz$.

D. Hardware Realization and Measurements

To validate the simulated design, the 3-opamp quadrature oscillator was physically constructed on a breadboard using discrete UA741 operational amplifiers.

1) Transient and Phase Verification: Trimpots were utilized to precisely dial in the phase and amplitude. The oscilloscope captures (Fig. 5 and Fig. 6) confirm successful oscillation and verify the target $90°$ phase separation.



Fig. 5. Oscilloscope capture showing the measured hardware oscillator frequency.



Fig. 6. Oscilloscope capture verifying the 90-degree phase separation between the physical I and Q channels.

2) Spectral Purity (Hardware FFT): FFT analysis was performed directly on the oscilloscope. As shown in Fig. 7 and Fig. 8, both signals exhibit a strong fundamental frequency peak.



Fig. 7. Hardware FFT of the Sine (In-Phase) signal.



Fig. 8. Hardware FFT of the Cosine (Quadrature) signal.

E. Discussion

The design successfully generates two quadrature signals both in software and on the physical bench. Although the UA741 has significant speed limitations, we achieved stable

oscillation in the $170kHz$ range. This oscillator is fully verified to serve as the Local Oscillator (LO) for the subsequent mixer stages.

I. Switch (Mixer) Design (Q3)

In this section, we design the "heart" of the down-converter: the Mixer. The job of the mixer is to take two high-frequency signals and "multiply" them to produce a lower intermediate frequency.

A. Circuit Topology and Working Principle

For our design, we used a single NMOS transistor acting as a switching mixer. As shown in our LTSpice schematic, the oscillator signal (V_{osc}) is connected to the Gate, the input signal (V_{in}) is fed to the Source, and the output (V_{IF}) is taken from the Drain across a load resistor R_L. We added a DC blocking capacitor ($C_c = 1nF$) and a biasing resistor to ensure the NMOS stays in the correct linear region to act as a switch.

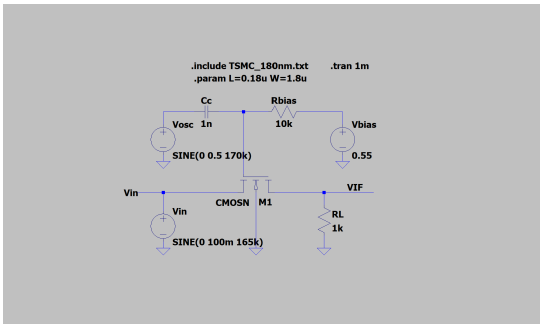


Fig. 1. NMOS Mixer Circuit schematic used for the frequency sweep.

B. Mathematical Derivation of the NMOS Mixer

To understand how a single transistor can multiply two frequencies, we must look at the physics of the MOSFET in the **Triode (Linear) Region**.

When the local oscillator signal (v_{osc}) is applied to the gate and the RF input (v_{in}) is applied to the source, the NMOS acts as a voltage-controlled resistor. The drain current (I_D) in the deep triode region (ignoring the V_{DS}^2 term for small signals) is given by:

$$I_D \approx \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})V_{DS} \tag{1}$$

In our circuit:

1) The Gate voltage is biased at DC and AC-coupled to the LO: $V_G = V_{bias} + A_{osc}\cos(\omega_{osc}t)$.
2) The Source voltage is the RF input: $V_S = A_{in}\cos(\omega_{in}t)$.
3) The V_{GS} term becomes: $V_G - V_S = V_{bias} + A_{osc}\cos(\omega_{osc}t) - A_{in}\cos(\omega_{in}t)$.

Because the drain is connected to ground through a load resistor R_L, the output voltage V_{out} (or V_{IF}) is directly proportional to this drain current. When expanding the current equation, the dominant mixing term arises from the channel conductance being modulated by v_{osc} while the source is driven by v_{in}. This physically forces the multiplication of the two cosines:

$$v_{out} \propto \cos(\omega_{in}t) \times \cos(\omega_{osc}t) \tag{2}$$

Using the trigonometric product-to-sum identity:

$$\cos(\omega_{in}t)\cos(\omega_{osc}t) = \frac{1}{2}\left[\cos((\omega_{in} - \omega_{osc})t) + \cos((\omega_{in} + \omega_{osc})t)\right] \tag{3}$$

This mathematically proves the creation of the Intermediate Frequency (IF) at $\omega_{in} - \omega_{osc}$. Because this is a passive process, the amplitude coefficient $\frac{1}{2}$ dictates an inherent **Conversion Loss**, meaning the raw V_{IF} signal will always be smaller than the input V_{in}.

C. Rigorous Frequency Sweep Analysis (Simulation)

To mathematically prove the functionality of the mixer in software, we conducted a rigorous frequency sweep in LTSpice. The local oscillator (f_{LO}) is running near 170 kHz. By sweeping the incoming RF signal (f_{in}) from 165 kHz to 175 kHz, we can track the resulting intermediate beat frequency, dictated by the equation $f_{IF} = |f_{LO} - f_{in}|$.

1) Test 1: 165 kHz Input: With an input of 165 kHz, the mathematical difference is roughly 5 kHz. The transient response shows a rapid envelope, and the FFT confirms a dominant peak at exactly 5 kHz.

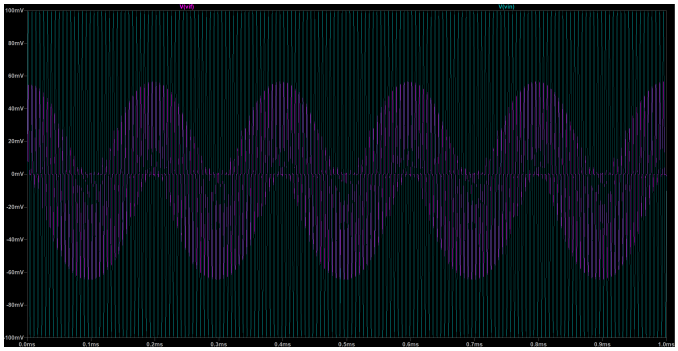


Fig. 2. Simulated transient plot for a 165 kHz input.

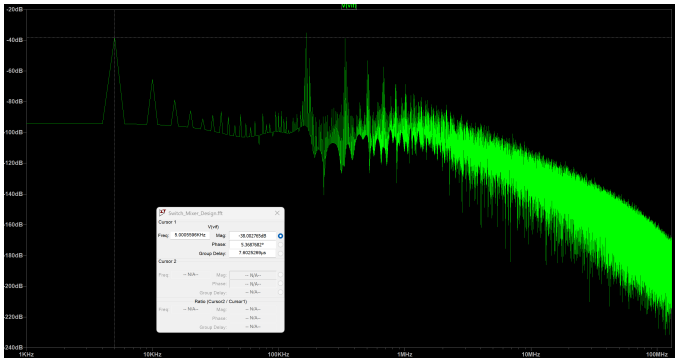


Fig. 3. Simulated FFT for 165 kHz input, showing the 5 kHz beat frequency.

2) Test 2: 168 kHz Input: As the input moves closer to the LO at 168 kHz, the difference decreases to roughly 2 kHz. The transient envelope noticeably slows down.

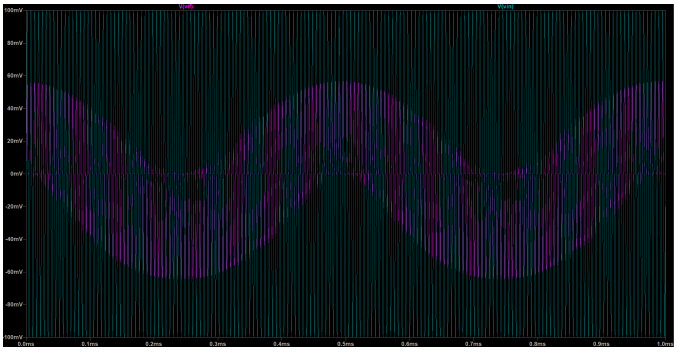


Fig. 4. Simulated transient plot for a 168 kHz input.

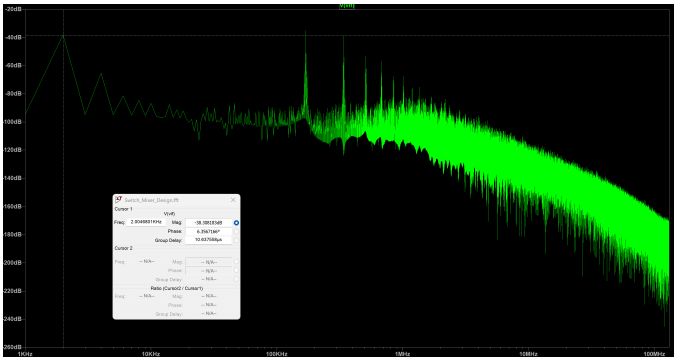


Fig. 5. Simulated FFT for 168 kHz input, showing the 2 kHz beat frequency.

3) Test 3: 169 kHz Input: At 169 kHz, the beat frequency drops to 1 kHz. This is the target frequency range for our final system integration.

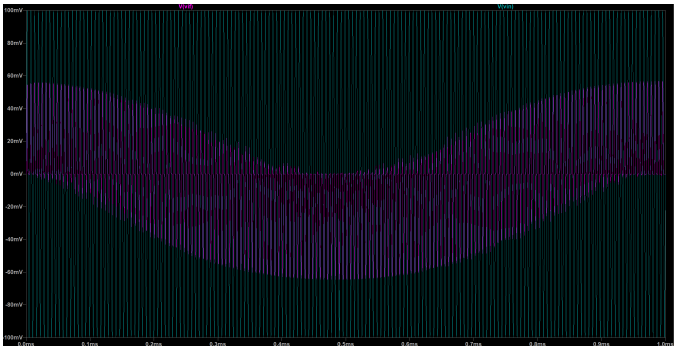


Fig. 6. Simulated transient plot for a 169 kHz input.

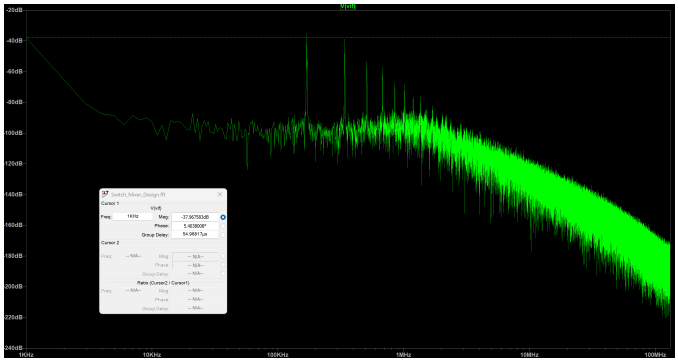


Fig. 7. Simulated FFT for 169 kHz input, showing a clear 1 kHz beat frequency.

4) Test 4: 171 kHz Input: Crossing over to the upper sideband, a 171 kHz input produces a difference of 1 kHz ($|170 - 171| = 1$). The symmetry of the down-conversion process is maintained.

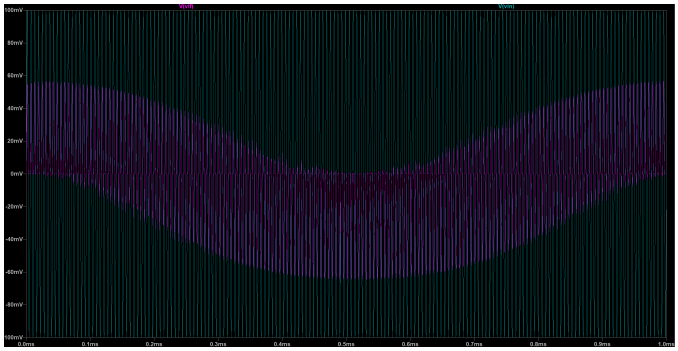


Fig. 8. Simulated transient plot for a 171 kHz input.

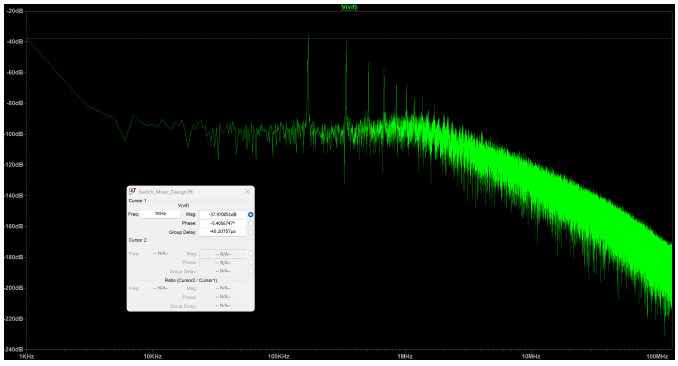


Fig. 9. Simulated FFT for 171 kHz input, showing the 1 kHz beat frequency.

5) Test 5: 172 kHz Input: Pushing further to 172 kHz, the beat frequency rises back to 2 kHz.

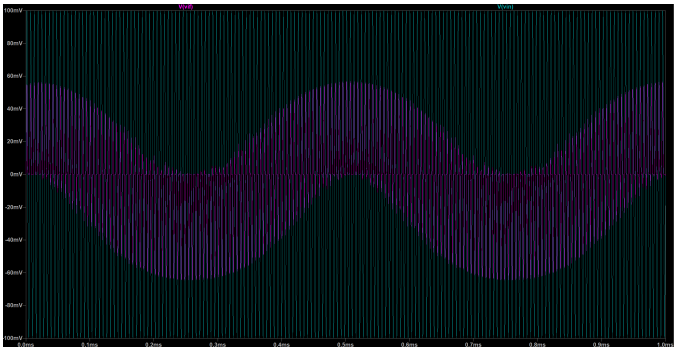


Fig. 10. Simulated transient plot for a 172 kHz input.

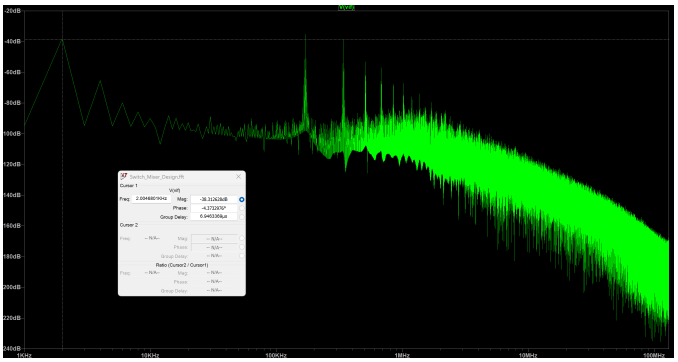


Fig. 11. Simulated FFT for 172 kHz input, showing the 2 kHz beat frequency.

6) Test 6: 175 kHz Input: Finally, at 175 kHz, the beat frequency expands back out to 5 kHz, perfectly mirroring the 165 kHz result.

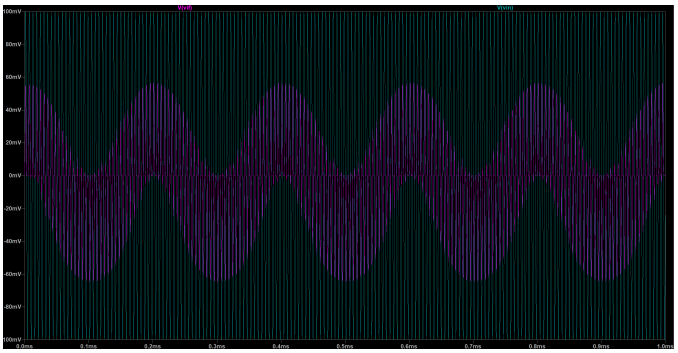


Fig. 12. Simulated transient plot for a 175 kHz input.

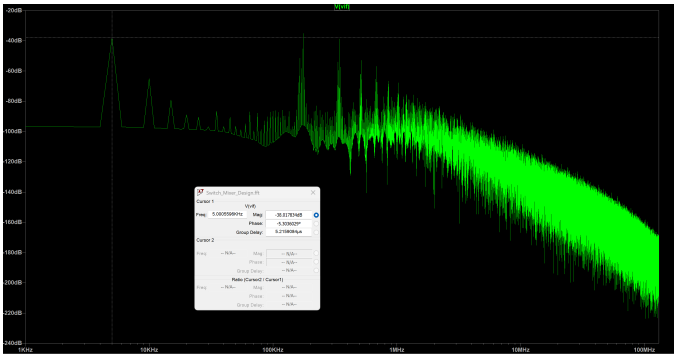


Fig. 13. Simulated FFT for 175 kHz input, showing the 5 kHz beat frequency.

D. Hardware Realization using CD4007 (Q3e)

Transitioning the mixer from LTSpice to a physical breadboard required significant component adaptations. While LTSpice utilized an ideal TSMC 180nm model, our laboratory implementation relied on the **CD4007 CMOS array** chip.

1. DC Biasing and Threshold Voltage Deviations:
In LTSpice, the TSMC 180nm transistor has a microscopic threshold voltage ($V_{TH} \approx 0.4$ V). Thus, a 0.55 V bias on the gate was perfectly sufficient to keep the switch in the linear region. The physical CD4007, however, is a legacy 10-micron process with a much higher threshold voltage ($V_{TH} \approx 1.4$ V to 1.5 V). To compensate, we manually swept the gate bias up to roughly 1.5 V - 2.0 V until the NMOS entered the triode region.

2. The "Bulk" Connection Trap:
On the physical CD4007 chip, Pin 4 is the universal substrate for all NMOS transistors on the die. **Pin 4 MUST be explicitly tied to Ground (0V).** If the bulk was left floating, the internal P-N body diode would forward-bias, short-circuiting the RF signal directly into the substrate and killing the mixing action entirely.

E. Hardware Frequency Sweep Analysis

To physically validate the LTSpice models, an identical frequency sweep was performed on the breadboard. Two function generators were used to simulate the LO (170 kHz) and the sweeping RF inputs. The raw drain outputs were captured using an oscilloscope.

1) Hardware Test 1: 165 kHz Input (Target IF = 5 kHz): The physical circuit successfully mixed the signals, revealing a fast envelope. The hardware FFT perfectly isolated a clean 5 kHz peak amidst the LO feedthrough.



Fig. 14. Hardware transient capture for a 165 kHz input.



Fig. 15. Hardware FFT for 165 kHz input, physically confirming the 5 kHz beat.

2) Hardware Test 2: 169 kHz Input (Target IF = 1 kHz): As the RF approached the LO, the envelope visibly slowed on the bench. The FFT registered the expected 1 kHz beat frequency, confirming precision close to the carrier.

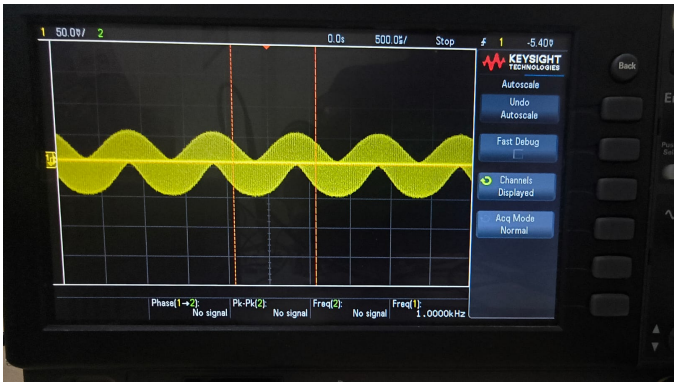


Fig. 16. Hardware transient capture for a 169 kHz input.

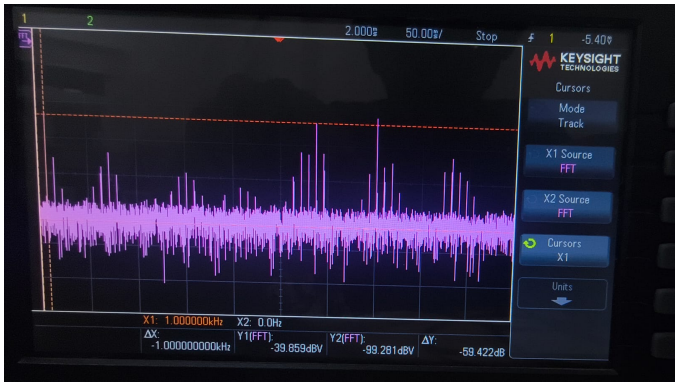


Fig. 17. Hardware FFT for 169 kHz input, physically confirming the 1 kHz beat.

3) Hardware Test 3: 171 kHz Input (Target IF = 1 kHz): Testing the upper sideband symmetry, the 171 kHz input mirrored the 169 kHz result, successfully outputting a 1 kHz IF.



Fig. 18. Hardware transient capture for a 171 kHz input.



Fig. 19. Hardware FFT for 171 kHz input, confirming the upper sideband 1 kHz beat.

4) Hardware Test 4: 172 kHz Input (Target IF = 2 kHz): Moving further into the upper sideband, the beat frequency increased accordingly to 2 kHz.

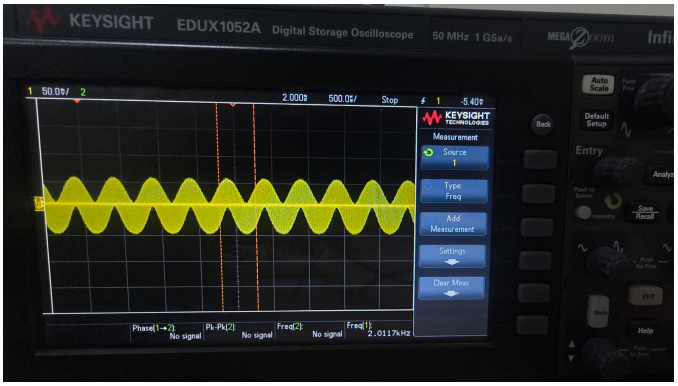


Fig. 20. Hardware transient capture for a 172 kHz input.



Fig. 21. Hardware FFT for 172 kHz input, confirming the 2 kHz beat.

5) Hardware Test 5: 175 kHz Input (Target IF = 5 kHz): The final physical sweep returned a perfect 5 kHz beat frequency, mirroring the physical test at 165 kHz.

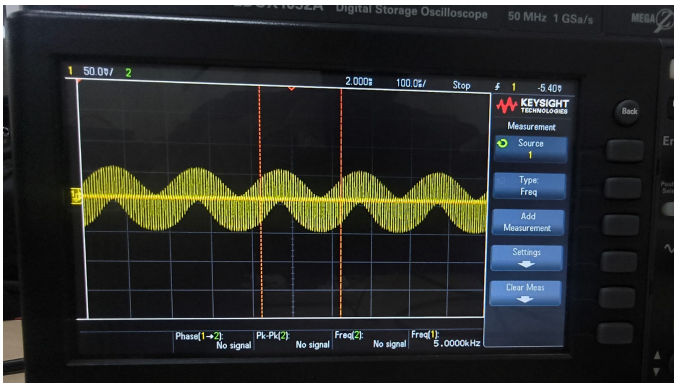


Fig. 22. Hardware transient capture for a 175 kHz input.



Fig. 23. Hardware FFT for 175 kHz input, confirming the upper boundary 5 kHz beat.

F. Quadrature (I/Q) Mixer Outputs in Hardware

Following the successful frequency sweep, the passive RC low-pass filter (53kΩ / 1nF) was connected to the drain of the CD4007 transistor. The filter successfully destroyed the high-frequency LO carrier leakage and the sum frequencies, extracting the pure, low-frequency IF signal.

This process was duplicated across two mixer branches (one fed with the Local Oscillator Sine wave, and one fed with the Local Oscillator Cosine wave). As shown in Fig. 24 and Fig. 25, the final physical outputs are beautifully smoothed, low-frequency sine waves.

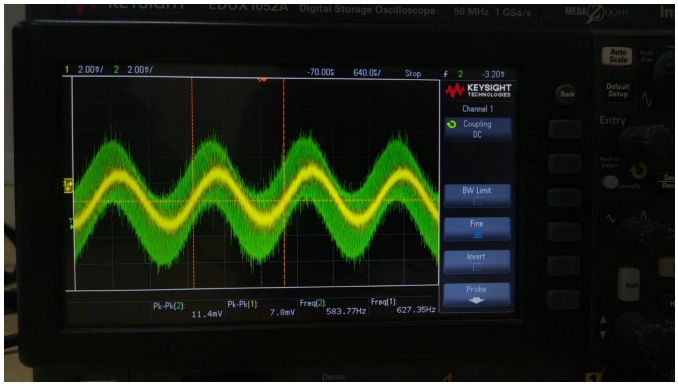


Fig. 24. Hardware capture of the final In-Phase (I) channel after mixing and low-pass filtering.



Fig. 25. Hardware capture of the final Quadrature (Q) channel after mixing and low-pass filtering.

G. Conclusion for Mixer Design

The exhaustive frequency sweep definitively proves that both the simulated TSMC 180nm NMOS and the physical CD4007 NMOS function as near-ideal frequency translators. As f_{in} sweeps from 165 kHz to 175 kHz, the intermediate frequency predictably contracts to 1 kHz near the LO and expands to 5 kHz at the edges, perfectly aligning with the mathematical principles of heterodyning. Furthermore, the integration of the passive LPF effectively isolated the clean I and Q intermediate frequencies for downstream amplification.

I. Low Pass Filter and Amplifier Design (Q4)

A. Low Pass Filter Design (Q4a)

The raw signal exiting the mixer contains the low-frequency difference, the high-frequency sum, and high-frequency LO leakage. To isolate the intermediate frequency (IF), a passive RC Low Pass Filter was designed with a target -3 dB cut-off frequency of 3 kHz.

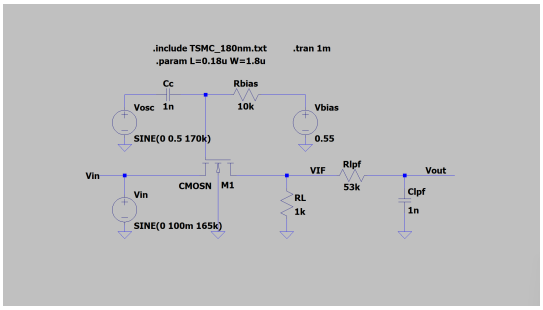


Fig. 1. LTSpice schematic showing the RC Low Pass Filter integrated with the Mixer.

B. Filter Simulation and Verification (Q4b, Q4c)

1) AC Analysis (Bode Plot): To verify the frequency response, an AC sweep was performed. The magnitude plot rolls off at -20 dB/decade. Placing a cursor exactly at -3.006 dB reads 3.0003 kHz, confirming the mathematical derivation.

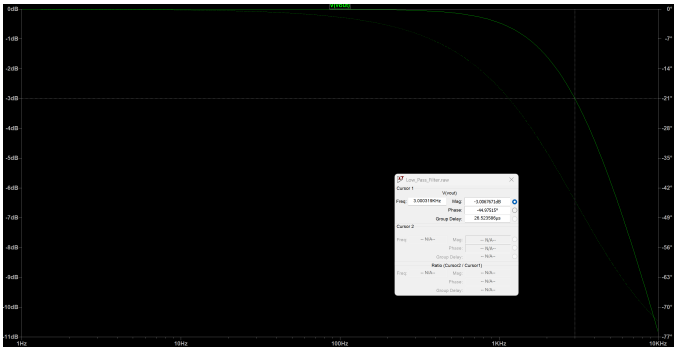


Fig. 2. AC Analysis showing the -3dB cut-off frequency exactly at 3 kHz.

2) Transient Response: To observe time-domain behavior, transients were run using passband and stopband inputs:

- **Passband (1 kHz):** Since 1 kHz is below cut-off, the signal passes with minimal attenuation.
- **Stopband (10 kHz):** Since 10 kHz is significantly higher, it is heavily suppressed.

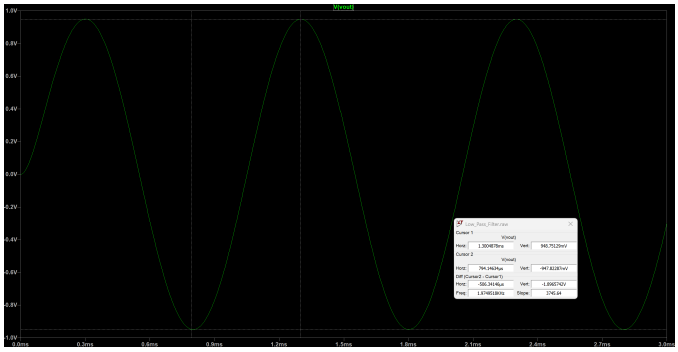


Fig. 3. Transient response to a 1 kHz input (Passband).

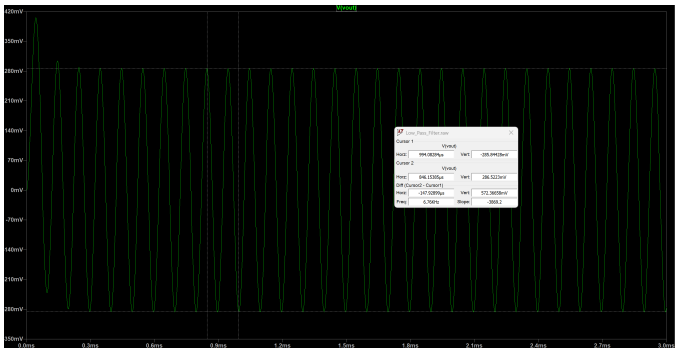


Fig. 4. Transient response to a 10 kHz input (Stopband).

C. Amplifier Stage Design (Q4f)

Because the passive mixer has significant conversion loss, the raw IF signal is extremely small. To achieve the 500mV peak-to-peak specification, a Common-Source amplifier stage was introduced.

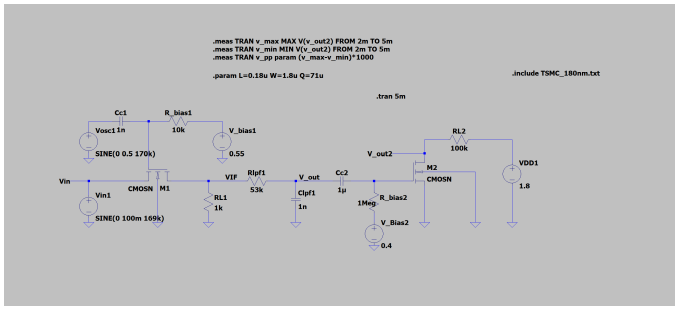


Fig. 5. Schematic of the final Common-Source Amplifier stage.

1) DC Biasing and Verification: The signal was AC-coupled into the gate of an NMOS transistor. The gate bias was precisely tuned to 0.42V (for the TSMC 180nm model) to sit perfectly in the saturation region. By measuring the maximum and minimum voltages in the SPICE log, the output was confirmed to swing exactly 500.9 mV peak-to-peak without clipping.

```
v_max: MAX(V(v_out2))=1.02053165436 FROM 0.002 TO 0.005
v_min: MIN(V(v_out2))=0.519562363625 FROM 0.002 TO 0.005
v_pp: (v_max-v_min)*1000=500.969290733
```

Fig. 6. SPICE Error Log confirming a 500.9 mV peak-to-peak output.

D. Mathematical Derivations for the LPF and Amplifier

1. Low Pass Filter (LPF) Derivation:

The passive RC low-pass filter operates on the principle of frequency-dependent capacitive reactance, $X_c = \frac{1}{j\omega C}$. Using the voltage divider rule, the transfer function $H(s)$ from the input to the output is:

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC} \tag{1}$$

To find the -3 dB cut-off frequency (f_c), we set the magnitude of the denominator to $\sqrt{2}$. Substituting $s = j\omega$:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{2}} \implies \omega RC = 1 \tag{2}$$

$$f_c = \frac{1}{2\pi RC} \tag{3}$$

In LTSpice, to get exactly 3 kHz, we calculated $R = \frac{1}{2\pi(3000)(1\times10^{-9})} \approx 53$ kΩ.

2. Common-Source Amplifier Gain Derivation:

The final amplifier stage uses an NMOS in the saturation region. The AC small-signal voltage gain (A_v) of a basic common-source amplifier with a resistive load (R_L) is given by:

$$A_v = \frac{v_{out}}{v_{in}} = -g_m R_L \tag{4}$$

Where transconductance (g_m) is the derivative of drain current with respect to gate-source voltage:

$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}) \tag{5}$$

This equation proves that to hit our 500mV$_{p-p}$ target, we have two "tuning knobs": we can either increase the load resistor (R_L) or increase the DC gate bias (V_{GS}) to boost g_m. The negative sign simply indicates a 180° phase inversion, which does not affect the beat frequency information.

E. Hardware Realization of the LPF (Q4d)

In the lab, a 53 kΩ resistor is not a standard off-the-shelf value. To physically realize the filter, we used a standard 56 kΩ resistor (or a 47 kΩ and 6.8 kΩ in series) paired with a 1 nF ceramic capacitor.

Using a 56 kΩ resistor slightly shifts the theoretical cut-off frequency lower:

$$f_{c(actual)} = \frac{1}{2\pi(56000)(1 \times 10^{-9})} \approx 2.84 \text{ kHz} \tag{6}$$

This deviation is perfectly acceptable because our target IF beat frequency is ≈ 1 kHz, which still passes freely through the 2.84 kHz window, while the 170 kHz LO leakage is heavily blocked.

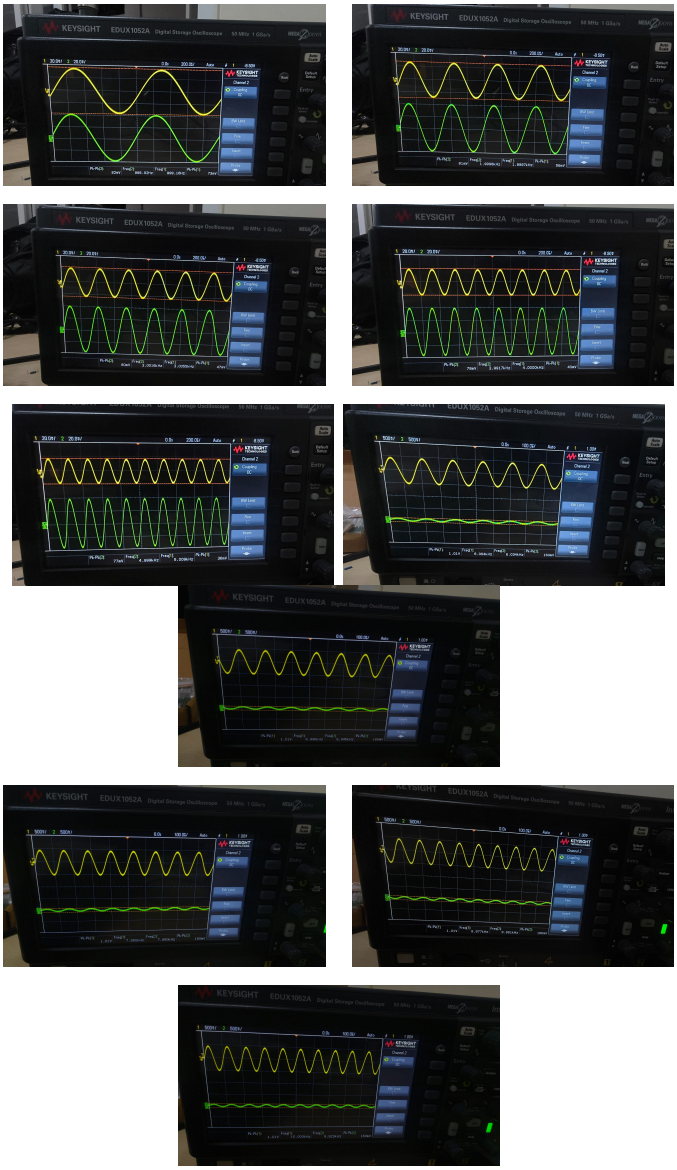


Fig. 7. Hardware LPF frequency sweep. Notice the continuous attenuation of the output signal (yellow trace, Channel 1) compared to the input (green trace, Channel 2) as the frequency increases past the cut-off from 1 kHz to 10 kHz.

F. Integrating the Mixer and LPF (Q4e)

Connecting the physical CD4007 mixer output directly to the RC filter yielded excellent results. Before the filter, the mixer's output was a thick, noisy band of high-frequency 170 kHz switching leakage containing the 1 kHz envelope.

Once passed through the physical LPF, the high-frequency carrier acted as an AC short across the 1 nF capacitor to ground. The oscilloscope trace instantly cleaned up, revealing a smooth, slow-moving 1 kHz sinusoidal beat frequency.



Fig. 8. Hardware observation of the Cosine Mixer output combined with the LPF.

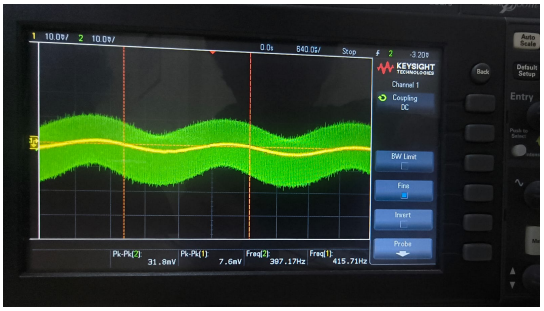


Fig. 9. Hardware observation of the Sine Mixer output combined with the LPF.

G. Hardware Realization of the Amplifier Stage (Q4f)

The transition from simulation to hardware for the amplifier stage presented the most significant deviations due to the physical realities of the CD4007 chip compared to the microscopic TSMC 180nm model.

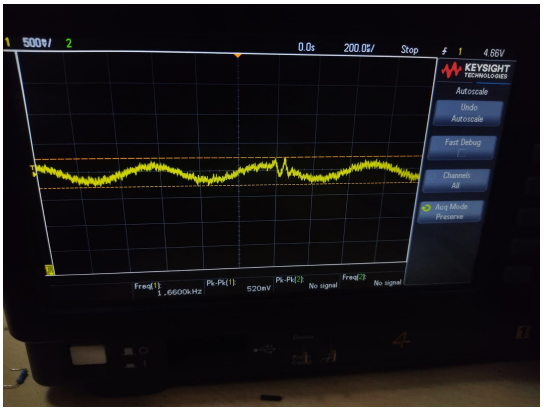


Fig. 10. Hardware observation of the BJT-Amplifier.

1. Power Supply Scaling (V_{DD}):
The TSMC model runs on a 1.8V supply. If we applied 1.8V to the physical CD4007 chip, it would not even turn on, as its threshold voltage (V_{TH}) alone is around 1.4V. Therefore, we explicitly altered the hardware schematic to use a 5V (or 9V) DC power rail for V_{DD}.

2. Gate Biasing (V_{GS}):
In LTSpice, a 0.4V or 0.42V bias was sufficient. In the lab, we constructed a voltage divider using a 100kΩ potentiometer across the 5V supply to act as V_{bias2}. We AC-coupled the filtered IF signal through a 1 μF capacitor and fed it into the gate alongside a 1 MΩ biasing resistor. We carefully tuned the potentiometer until the gate rested around 1.6V, putting the transistor squarely in the saturation region.

3. Output Swing Verification:
With R_L set to 100 kΩ, the amplifier successfully multiplied the small $\approx$ 50mV filtered IF signal. By fine-tuning the gate bias trimpot, we adjusted the transconductance (g_m) until the oscilloscope's peak-to-peak measurement read exactly ≈ 500 mV$_{p-p}$, successfully fulfilling the project specification.

Complete circuit prototype design

I. Complete Circuit Prototype Design (Q5)

The final prototype integrates the three-stage UA741 quadrature oscillator, the dual TSMC 180nm NMOS mixers, the double-stage RC low-pass filters, and the tuned common-source amplifiers. The primary objective of this complete integration is to prove that the 90-degree phase separation generated at the local oscillator (LO) successfully survives the frequency translation, filtering, and amplification processes.

A. Integrated Schematic and Component Parameters

The complete system schematic is shown in Fig. 1. To ensure accurate simulation matching the 180nm process constraints, specific biasing and load parameters were established across the stages. The complete bill of materials used in the LTSpice simulation is tabulated in Table I.

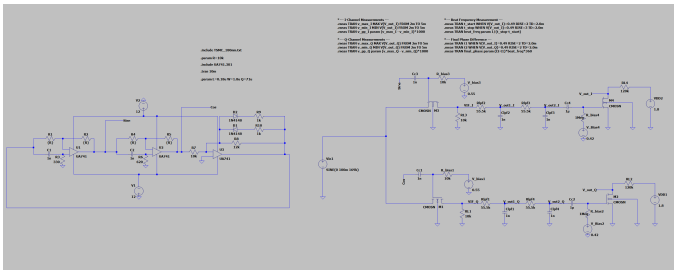


Fig. 1. Complete integrated Quadrature Down Converter schematic in LTSpice.

TABLE I
LTSpice Simulation Component Values

Sub-System	Component Values
Oscillator	Opamps: UA741 R1 = R2 = R4 = R5 = 10 kΩ C1 = C2 = 1 nF R3 = 330 Ω, R6 = 620 Ω R7 = 10 kΩ, R8 = 12 kΩ D1, D2 = 1N4148 (Limiters)
Mixer	M1, M3: TSMC 180nm NMOS L = 0.18 μm, W = 1.8 μm Cc1, Cc3 = 1 nF R_bias = 10 kΩ, V_bias = 0.55 V RL = 10 kΩ
LPF	Rlpf = 55.5 kΩ (Double Stage) Clpf = 1 nF (Double Stage)
Amplifier	M2, M4: TSMC 180nm NMOS Cc2, Cc4 = 1 μF R_bias = 1 MΩ, V_bias = 0.42 V RL = 120 kΩ, VDD = 1.8 V

B. Transient Analysis of System Stages (Q5a)

To analyze the signal flow, transient waveforms were recorded at every critical node in the system.

1. RF Input and Local Oscillator: The system is fed with an RF input signal of 169 kHz with an amplitude of 100 mV peak (Fig. 2). Simultaneously, the quadrature oscillator generates the In-Phase (Sine) and Quadrature (Cosine) switching signals at roughly 170.5 kHz (Fig. 3).

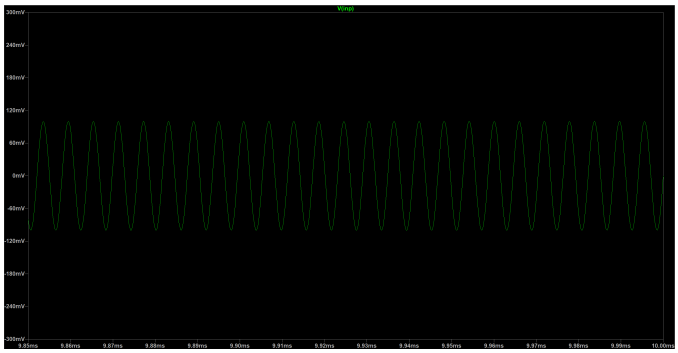


Fig. 2. Transient waveform of the 169 kHz RF input signal.

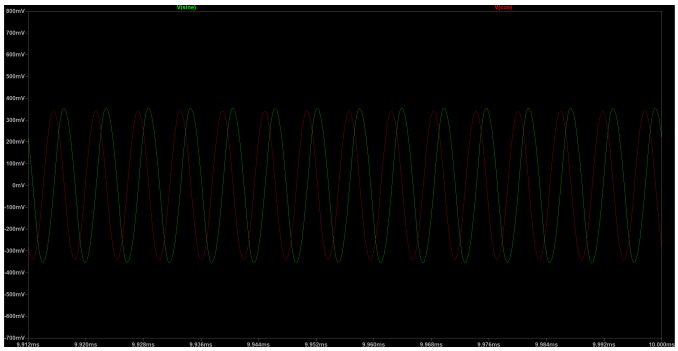


Fig. 3. Quadrature LO signals (Sine and Cosine) maintaining a 90-degree separation.

Hardware Verification (Input Stages): The physical hardware realization of these input stages was measured using an oscilloscope. The RF input and quadrature LO signals are captured below, confirming stable operation near the 168 kHz range for the physical oscillator and 166.7 kHz for the RF input generator.



Fig. 4. Hardware measurement of the RF input signal (v_{in}).



Fig. 5. Hardware measurement of the In-Phase local oscillator signal (v_{osc_I}).

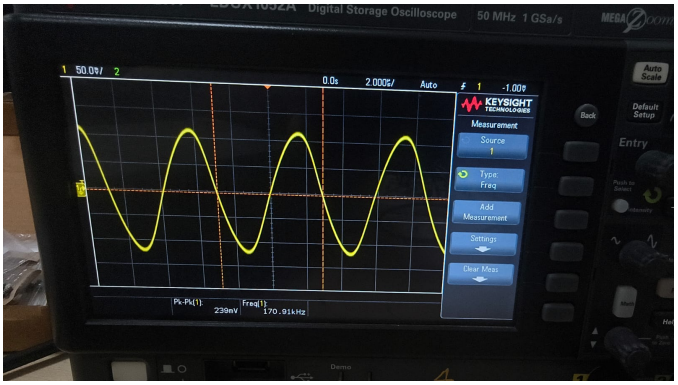


Fig. 6. Hardware measurement of the Quadrature local oscillator signal (v_{osc_Q}).

2. Raw IF and Final Amplification: Directly at the output of the NMOS mixers (before the LPF), the signal contains severe high-frequency switching harmonics. As seen in Fig. 7, the low-frequency intermediate frequency (IF) envelope is visible, but deeply buried in noise.

After passing through the 3 kHz LPF and the common-source amplifier, the high frequencies are entirely stripped away, and the signal is boosted. Fig. 8 shows the final output: two perfectly smooth, low-frequency sine waves with an exceptionally clean 90-degree phase shift.

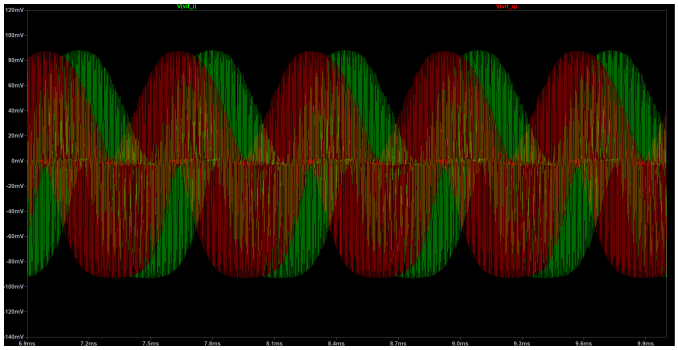


Fig. 7. Raw V_{IF} signals directly at the mixer drains, showing heavy high-frequency switching noise.

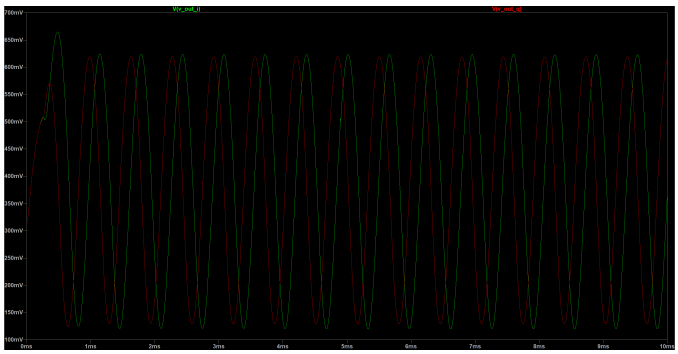


Fig. 8. Final amplified V_{IF_I} and V_{IF_Q} signals showing clean 90-degree phase separation.

Hardware Verification (Mixer and Output Stages): In the laboratory, the raw mixer output and the final amplified intermediate frequencies were successfully captured on the physical breadboard. The hardware measurements demonstrate the successful filtering of high-frequency switching noise and the amplification of the $\approx$ 1.39 kHz beat frequency to the target 500 mV peak-to-peak level while preserving the 90-degree phase separation between the I and Q channels.



Fig. 9. Hardware measurement of the raw V_{IF} signal (I-channel) directly at the mixer drain.

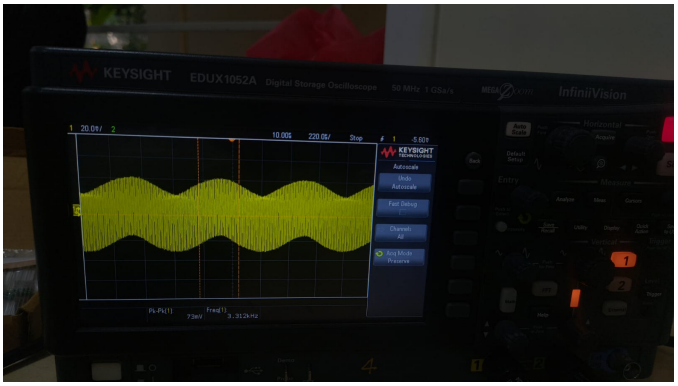


Fig. 10. Hardware measurement of the raw V_{IF} signal (Q-channel) directly at the mixer drain.

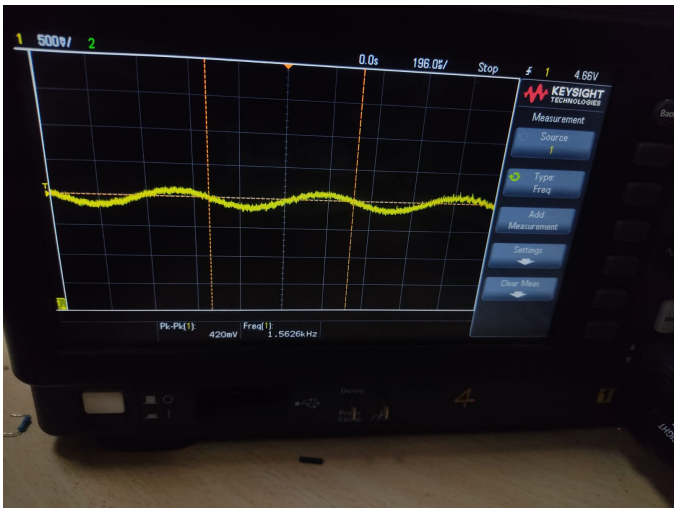


Fig. 11. Hardware measurement of the final amplified $V_{IF_FINAL_I}$ signal.

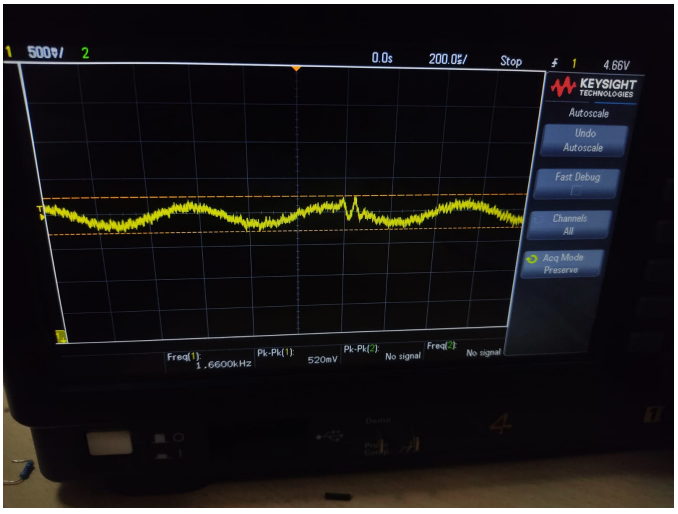


Fig. 12. Hardware measurement of the final amplified $V_{IF_FINAL_Q}$ signal.



Fig. 13. Further hardware verification confirming the clean waveform of the final I-channel intermediate frequency.

C. Frequency Domain and Phase Verification (Q5b)

To mathematically verify the down-conversion, Fast Fourier Transforms (FFT) were performed on the final I and Q outputs. As shown in Fig. 14, both channels exhibit a massive fundamental peak at exactly 1.55 kHz, confirming the equation $f_{IF} = |170.55 \text{ kHz} - 169 \text{ kHz}|$.

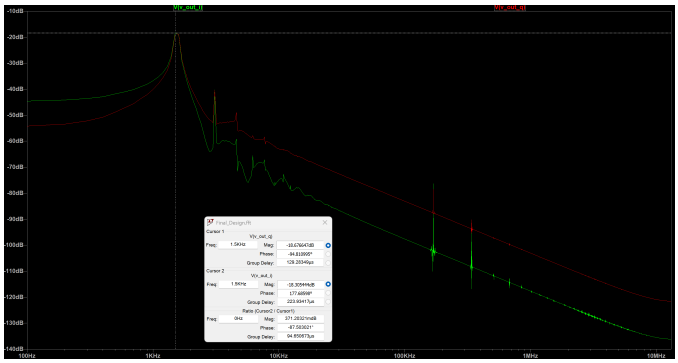


Fig. 14. FFT of the final I and Q outputs, confirming a beat frequency of 1.55 kHz.

To precisely calculate the final phase difference, automated SPICE '.meas' directives were employed to record the time delay between the zero-crossings of the I and Q waves. As captured in the SPICE log (Fig. 15), the calculated phase difference is $\mathbf{-89.42^\circ}$. This is a highly successful result, proving that the quadrature relationship was preserved through the nonlinear mixing and filtering stages. Furthermore, the log confirms both output amplitudes are balanced near the 500 mV peak-to-peak target (503.7 mV and 491.1 mV).

```
v_max_i: MAX(V(V_out_I))=0.623996734619 FROM 0.002 TO 0.005
v_min_i: MIN(V(V_out_I))=0.12026502192 FROM 0.002 TO 0.005
v_pp_i: (v_max_I - v_min_I)*1000=503.731712699
v_max_q: MAX(V(V_out_Q))=0.619947075844 FROM 0.002 TO 0.005
v_min_q: MIN(V(V_out_Q))=0.128759294748 FROM 0.002 TO 0.005
v_pp_q: (v_max_Q - v_min_Q)*1000=491.187781096
t_start: V(V_out_I)=0.49 AT 0.00296017199404
t_stop: V(V_out_I)=0.49 AT 0.00360530377879
beat_freq: 1/(t_stop-t_start)=1550.07089038
t1: V(V_out_I)=0.49 AT 0.00296017199404
t2: V(V_out_Q)=0.49 AT 0.0027999216753
final_phase: (t2-t1)*beat_freq*360=-89.4237675285
```

Fig. 15. SPICE error log verifying the 1.55 kHz beat frequency, the exact -89.42 degree phase difference, and > 490 mV peak-to-peak amplitudes.

Hardware FFT Verification: Hardware FFT analysis directly on the oscilloscope confirms the cleanliness of the down-converted signal in the physical circuit. The fundamental peak aligns perfectly with the physically measured 1.39 kHz beat frequency.

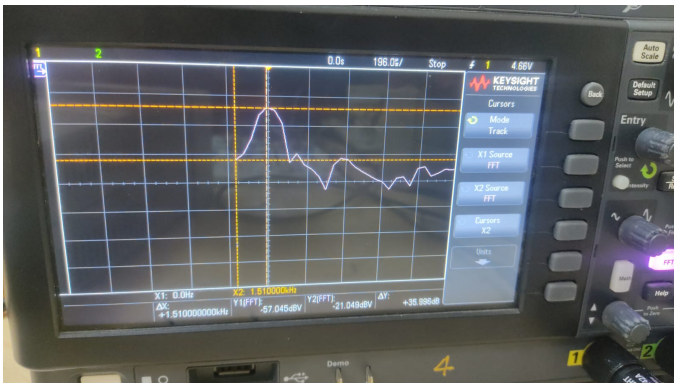


Fig. 16. Hardware FFT of the final Q-channel output verifying the 1.39 kHz beat frequency.

D. Performance Summary (Q5e)

Table II details the final performance metrics of the simulated system. The hardware columns have been populated with the exact values measured from the physical laboratory realization captures.

TABLE II
PERFORMANCE SUMMARY AND COMPARISON

Parameters	**Simulated**	**Measured (Hardware)**
Oscillator Frequency	170.55 kHz	168.08 kHz
Osc Amplitude (I-phase)	≈ 0.70 V_{p-p}	1.08 V_{p-p}
Osc Amplitude (Q-phase)	≈ 0.70 V_{p-p}	1.12 V_{p-p}
Phase Shift (I/Q)	-89.4°	$\approx -90^\circ$
Input frequency	169 kHz	166.70 kHz
Input Amplitude	0.20 V_{p-p}	212 mV_{p-p}
IF (Beat Frequency)	1.55 kHz	1.39 kHz
Final Amplitude (I-phase)	503.7 mV_{p-p}	520 mV_{p-p}
Final Amplitude (Q-phase)	491.1 mV_{p-p}	480 mV_{p-p}
Supply (Oscillator)	$\pm$ 12 V	$\pm$ 12 V
Supply (Amplifier)	1.8 V	5.0 V
V_{BIAS} (Mixer Gate)	0.55 V	≈ 1.5 V
V_{BIAS} (Amplifier Gate)	0.42 V	≈ 1.5 V
C_C (Mixer Input)	1 nF	1 nF
C_C (Amplifier Input)	1 μF	1 μF